

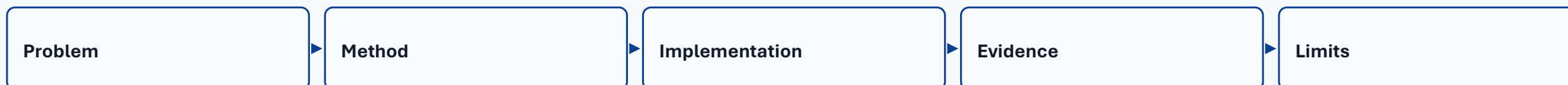


Development of an Intelligent Assistant for Data Preparation Automation in Urban Transportation Management Systems

Flowmatic master's thesis defence, 2026 | School of Computer Science and Engineering

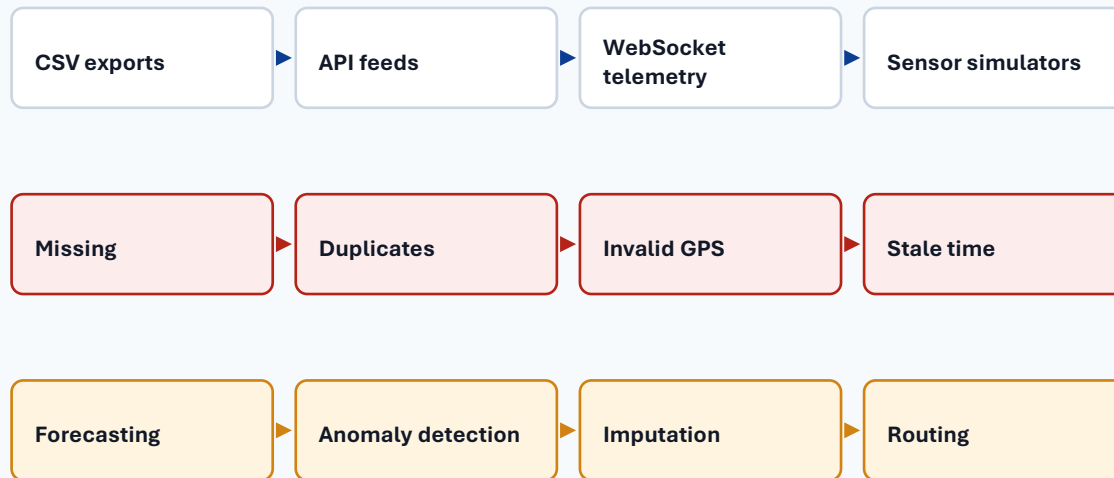
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Dirty transport data breaks the path from operations to analytics.

The research problem is not just missing values; it is the missing auditable preparation-to-inference chain.



Research problem

Urban-transport preparation is often performed outside the model workflow, leaving weak lineage, inconsistent quality measurement, and unclear downstream impact.

The gap is measurement linkage, not a claim that no tools exist.

Existing systems solve parts of the workflow; the thesis evaluates the chain end to end within a prototype.

Existing solutions	Typical limit	Research gap	Flowmatic contribution
Scripts / notebooks	Hard to audit and replay	No shared evidence trail	DQI + cleaning + export records
ETL / MLOps tools	Model and prep tracked apart	Weak transport-specific validation	Preparation tied to model registry
ITS analytics	Models evaluated separately	Unclear impact of data defects	Stress tests + ablation + routing check

Contribution boundary: Flowmatic does not claim a new universal neural architecture, new DQI theory, or production-certified ITS deployment.

The aim is implemented through six objectives and four scoped questions.

Each research question is tied to measurable evidence and a visible limitation.

Aim

Develop and evaluate an intelligent software platform for automated transport-data preparation with DQI, reproducible preprocessing, and downstream model integration.

Object

Data preparation process in urban transportation management systems.

Subject

Software mechanisms for quality assessment, preprocessing, routing, and model-ready data preparation.

RQ1

DQI recovery under corruption

RQ2

Batch and streaming prototype operation

RQ3

Model portfolio vs simple baselines

RQ4

Rule-router policy conformance

The scientific claim is the evaluated chain, not the existence of the platform.

Prior publications and standard components are separated from thesis-new evidence.

Scientific contribution

Linked evaluation protocol: DQI recovery, downstream-model effect, routing conformance, and reproducibility artifacts.

Engineering contribution

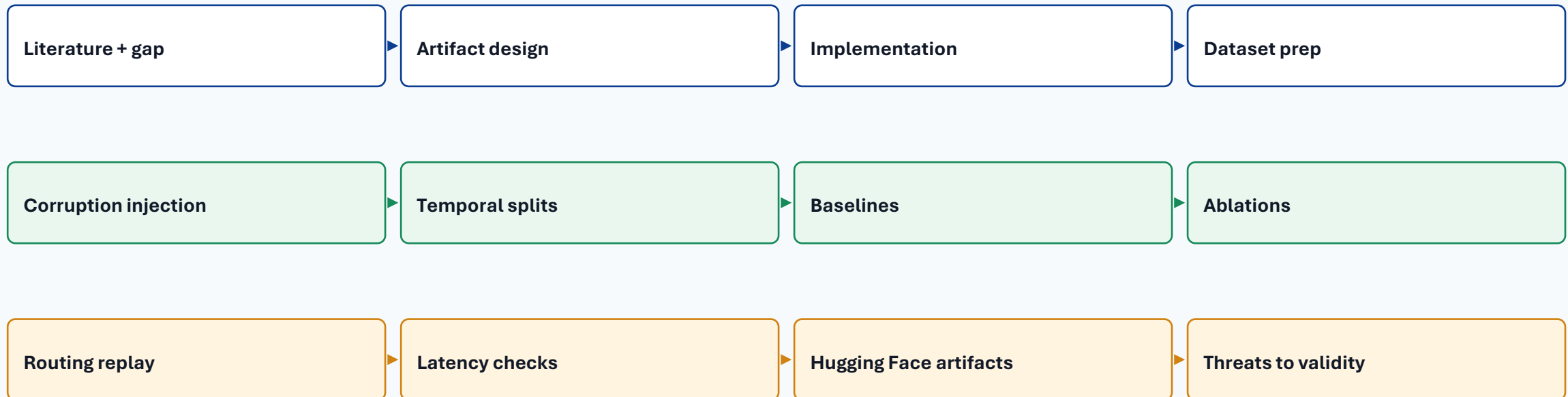
Containerised Flowmatic prototype: Vue, NestJS, PostgreSQL, MinIO, RabbitMQ, simulators, inference service.

Thesis-new evidence

Corruption stress test, baseline tables, preparation ablation, 140-event routing check, explicit limitation ledger.

Not claimed: new neural architecture, learned router, universal DQI theory, or live municipal production deployment.

The research workflow tests the artefact under controlled, reproducible conditions.



Falsifiable centre: if preparation does not improve quality or downstream behavior under controlled defects, the main thesis claim weakens.



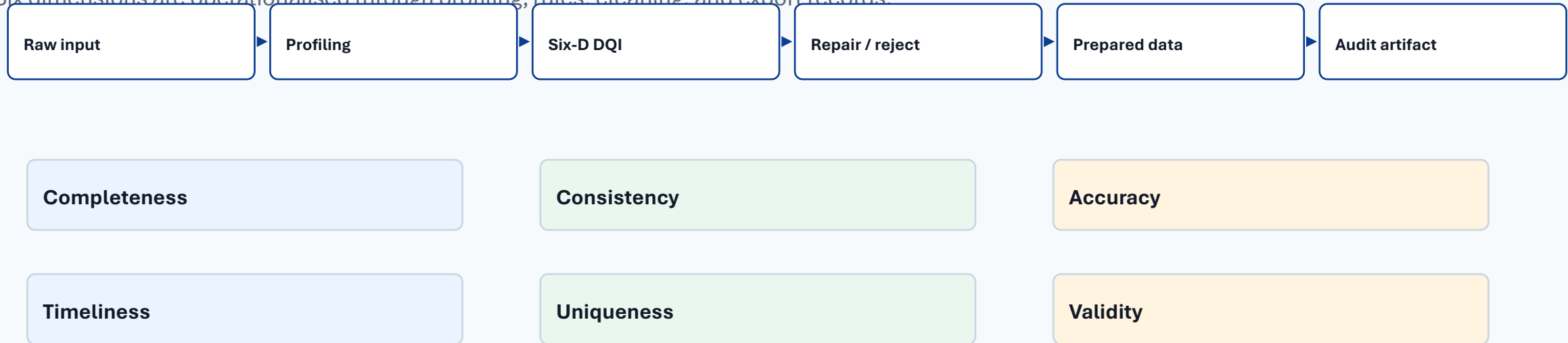
The evidence is reproducible, but not all data are externally real-world validated.

This slide makes semi-synthetic and simulated data visible before results appear

Dataset	Provenance	Scale	Use	Limit
Astana traffic	NDA-derived semi-synthetic	30,000 rows	Batch API, stress, forecasting	Reproducible CSV; source calibration limited
HF weather / ETT / traffic	Public bundles	Offline windows	Neural training	Not all re-uploaded through Nest API
Simulator events	Generated HTTP/WebSocket	140 routing events	Auto routing check	Policy-aligned labels
Injected defects/anomalies	Controlled perturbation	0-20% corruption	Stress / TranAD AUROC	Not municipal incident labels

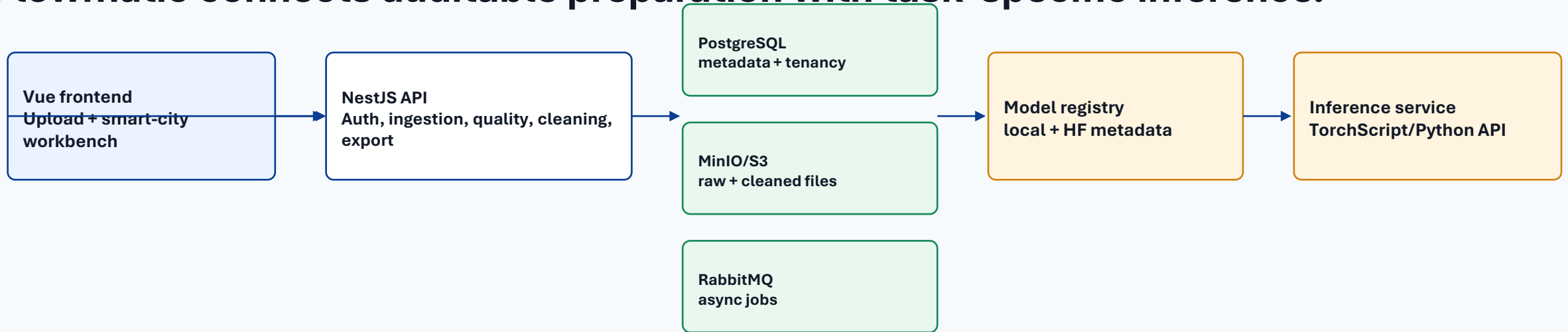
DQI makes preparation auditable before data reach models.

Six dimensions are operationalised through profiling, rules, cleaning, and export records.



Important boundary: at 20% corruption, quality recovery includes row rejection; it is not purely in-place imputation.

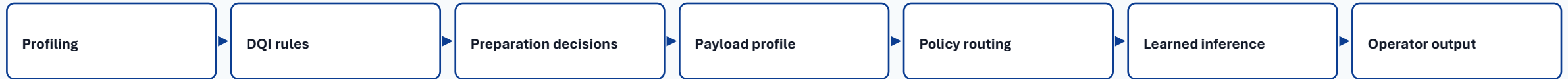
Flowmatic connects auditable preparation with task-specific inference.



Optional / not central evaluation: LLM narration and federated coordinator demo. They are prototype features, not the basis of the scientific claim.

The system is intelligent as a bounded hybrid assistant, not as general AI.

Automation combines deterministic diagnostics, policy routing, learned models, and operator control.



Deterministic automation
schema checks, missing values, duplicates,
ranges, stale records

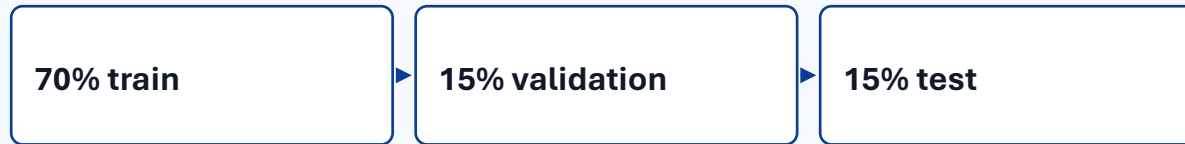
Policy decisions
Manual or Auto routing through metadata and
priority rules

Task-specific learning
forecasting, anomaly, imputation, severity
classification

LLM narration is optional and not used as the scientific basis of the thesis.

The experiment design separates leakage control from statistical strength.

Temporal hold-out is strong; three-seed variability should be read descriptively.



Leakage controls

Temporal split, scalars fitted on train, fixed test tail, identical partition for baselines.

Seeds
42, 7, 2026

Metrics
RMSE, macro-F1, AUROC, masked
MSE

Caution
No broad significance claim from n=3

Preparation recovered measurable quality, partly by rejecting invalid records.

At 20% corruption, DQI improved from 0.788 to 0.833 while 987 rows were dropped.



0.788

Before

20% corruption

0.833

After

cleaning

987

Rows dropped

9.87% of 10,000

Interpretation

Quality recovery is measurable, but the mechanism is auditably conservative: reject or repair defective records.

Downstream results support capability, with explicit task limits.

Metrics are separated by task to avoid pretending RMSE, F1, AUROC, and MSE are comparable.

Task	Model	Headline	Scope note
Forecast	PatchTST density	0.561 RMSE	vs naive z-RMSE 1.063
Classification	Transformer severity	0.911 macro-F1	rule-based labels
Anomaly	TranAD	0.841 AUROC	injected point anomalies
Imputation	SAITS	0.640 masked MSE	vs mean baseline 1.000

Defence framing: the models certify task slots inside Flowmatic; they do not establish a new forecasting or classification algorithm.

Preparation improved forecasting under a controlled upper-bound experiment.

PatchTST RMSE dropped from 0.9045 to 0.5802 after oracle-style restoration on 20% corrupted data.



0.3243

RMSE delta

lower is better

20%

Corruption

controlled input

What it proves

Restoring lost information improves downstream prediction under controlled corruption.

What it does not prove

A deployable imputation method will recover the same amount on live municipal data.

The prototype works, but the measured boundary matters.

Use core processing and inference latency numbers without turning them into production SLA claims

406 ms

Batch processing
30,000-row Astana export

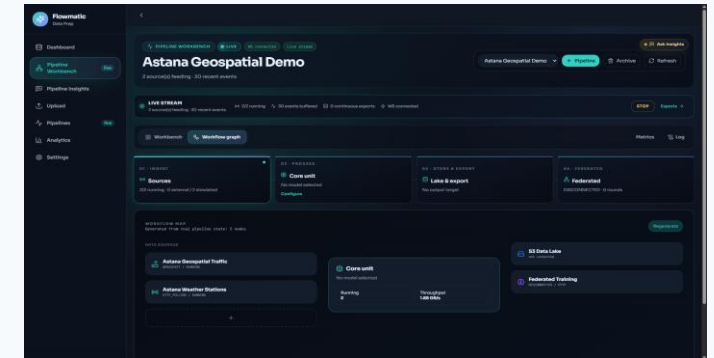
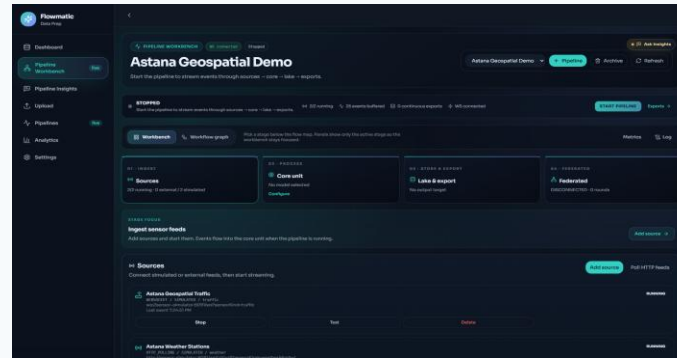
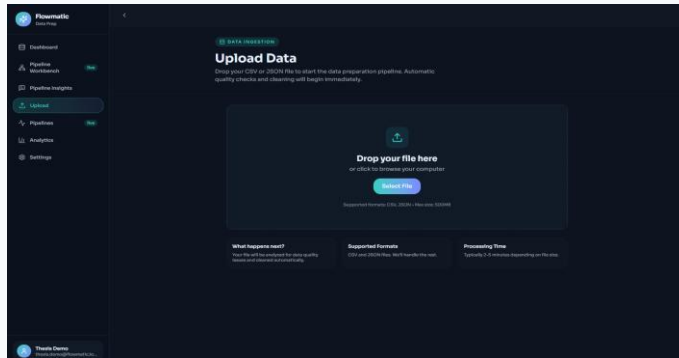
<2 ms

Compact inference
single-event benchmark

Docker

Prototype stack
local Compose topology

Not claimed
No HA, SLA, penetration test, formal municipal
deployment, or concurrency p95/p99 study.



Each research question has evidence and an explicit scope limit.

RQ	Evidence	Outcome	Scope limit
RQ1	DQI before/after stress test	Supported under corruption	clean files show ceiling effect
RQ2	406 ms batch + smart-city demo	Supported for prototype	Astana API validation only
RQ3	baselines, portfolio, ablations	Supported by task	limited baselines / n=3
RQ4	140-event routing replay	Supported as conformance	not expert-labelled accuracy

My contribution is the integrated implementation and thesis-new evaluation package.

Prior publications are supporting context; thesis claims are tied to new or extended artefacts

Personal thesis work

Architecture integration,
backend/frontend modules, data-quality
and routing logic, experiment scripts,
evidence audit, thesis narrative.

Shared / prior work

Co-authored publications, standard
neural architectures, open-source
frameworks, and source data constraints.

Thesis-new evidence

Stress tests, baseline comparisons,
ablation, routing-conformance
evaluation, limitation framing,
reproducibility pack.

Defence line: the publications show continuity; the thesis is defended through the new evaluated preparation-to-inference evidence chain.

CONCLUSION

Flowmatic is an evaluated research prototype for auditable transport-data preparation.

The strongest claim is narrow and defensible: controlled preparation evidence can be linked to downstream model behaviour

Achieved

Working prototype, DQI workflow, model registry, inference service, reproducible experiments.

Evidence supports

Quality recovery under controlled defects, model-slot capability, and policy-conformant routing.

Remaining work

Real external data, independent labels, deployable imputation, operator study, security/scale tests.

Core statement: Flowmatic makes preparation measurable, reproducible, and connected to downstream analytics - with limitations visible rather than hidden.

Backup slides for commission questions

Definitions, reconciliations, evidence tables, architecture limits, and exact numbers.

DQI definitions and formula

Pre-empts: why these dimensions and why equal weights.

Composite DQI = $\sum(w_i Q_i) / \sum(w_i)$, Q_i in $[0,1]$. Default weights are equal. Dimensions: completeness, consistency, accuracy, timeliness, uniqueness, validity.

Pre-empts: why these dimensions and why equal weights.

Source: Thesis Sec. 3.6

Clean 100/100 vs stress-test 0.833

Pre-empts: DQI inconsistency question.

Clean batch 100/100 is the platform quality score for a pristine 30,000-row upload. Stress-test 0.833 is the composite formula where timeliness is zero in that generated sample.

Pre-empts: DQI inconsistency question.

Source: Table 4.1; Table 5.9; evidence JSON

Dataset facts

Pre-empts: why trust semi-synthetic data.

Astana: 30,000 semi-synthetic NDA-derived rows. HF weather, ETT, traffic graph: offline neural bundles. Simulator: HTTP/WebSocket events. Labels and anomalies are generated or injected where stated.

Pre-empts: why trust semi-synthetic data.

Source: Appendix C; Sec. 5.2.1

Split, leakage, uncertainty

Pre-empts: n=3 CI and leakage questions.

Temporal 70/15/15 split; scalars fit on training data; three seeds 42, 7, 2026. Treat +/- values as descriptive multi-seed variability, not strong significance testing.

Pre-empts: n=3 CI and leakage questions.

Source: Sec. 5.2.2

Baseline table

Pre-empts: compared to what?

Naive z-RMSE 1.063 vs PatchTST 0.5609; majority 0.317, logistic 0.449, random forest 0.627, Transformer 0.911; mean imputation 1.000 vs SAITS 0.640.

Pre-empts: compared to what?

Source: `generated_baselines.tex`

Neural portfolio

Pre-empts: model zoo and registry questions.

Seven primary slots plus auxiliary iTransformer: PatchTST, TranAD, TimesBlock, STGCN, SAITS, Transformer classifier, DLinear, iTransformer speed slot.

Pre-empts: model zoo and registry questions.

Source: `production_portfolio.json`

iTransformer interpretation

Pre-empts: why keep weak model.

Level-speed RMSE is near 1.0; first-difference retune reaches 0.7586. The slot is retained as auxiliary/deployment parity, not as a headline success.

Pre-empts: why keep weak model.

Source: [generated_itransformer_retrain.tex](#)

Routing scope

Pre-empts: is 100% circular?

140 simulator events: 7 scenarios x 20 events. 100% policy conformance checks implementation consistency against documented scenario oracles; not optimal routing accuracy.

Pre-empts: is 100% circular?

Source: `generated_tables.tex`

Oracle restoration ablation

Pre-empts: did you use ground truth?

20% corrupted Astana density, PatchTST RMSE 0.9045 -> 0.5802. This is an upper-bound controlled information-recovery experiment, not deployable imputation proof.

Pre-empts: did you use ground truth?

Source: generated_prep_ablation.tex

Prior publication vs thesis-new

Pre-empts: overlap with [15].

Prior work: concept/platform continuity and co-authored publications. Thesis-new: stress tests, baselines, ablation, 140-event routing check, reproducibility and limitation framing.

Pre-empts: overlap with [15].

Source: Appendix A; Sec. 2.13.1

Personal contribution matrix

Pre-empts: what exactly did you do?

Own work: integration, implementation, scripts, experiments, analysis, defence framing. Shared: supervisor guidance, co-authored papers, public models/libraries, NDA-derived data source.

Pre-empts: what exactly did you do?

Source: Appendix A; repository evidence

Technology roles

Pre-empts: why this stack?

Vue for workbench; NestJS for API; PostgreSQL for metadata; MinIO for object storage; RabbitMQ for jobs; Python service for TorchScript inference; HF for model publication.

Pre-empts: why this stack?

Source: docker-compose.yml; Prisma schema



Codebase/module map

Pre-empts: walk through the code.

Backend: auth, ingestion, quality, cleaning, export, smart-city, registry/router. **Frontend:** upload, connectors, dashboard, insights. **Services:** simulator, inference, federated demo.

Pre-empts: walk through the code.

Source: repo map



Failure handling limits

Pre-empts: production architecture attack.

Current prototype has queues and storage, but no full DLQ/retry/idempotency proof, no p95/p99 concurrency benchmark, and preview loading can buffer large result files.

Pre-empts: production architecture attack.

Source: code audit

Security and production roadmap

Pre-empts: production readiness.

Current auth and org scoping exist. Needed before municipal pilot: threat model, upload sandboxing, model artifact validation, secret governance, HA, backup/restore, audit logs.

Pre-empts: production readiness.

Source: Prisma schema; security middleware

Reproducibility and exact numbers

Pre-empts: exact-number questions.

Exact numbers to remember: 30,000 rows; 406 ms; DQI +0.045 at 20%; 987 rows dropped; PatchTST 0.5609; F1 0.911; AUROC 0.8411; routing 140/140.

Pre-empts: exact-number questions.

Source: evidence JSON; tables